Submission CALC1-FINAL-SUB01

Question CALC1-FINAL-Q01
scratch: show step
1. $f'(x)=2x(3x-1)+(x^2+2)3$
2. $f'(3)=6(8)+(11)3=15$
=> FINAL: 15 (check units/interpretation)

Question CALC1-FINAL-Q02
scratch: show step
1. $\int 3x\,dx = (3/2)x^2$
2. $[3/2\ x^2]_0^5 = 37.5$
=> FINAL: 37.5 (check units/interpretation)

Question CALC1-FINAL-Q03
scratch: show step
1. $2w+2l=20 \Rightarrow l=10-w$
2. $A(w)=w(10-w)$; at $w=7$, $A=21$
3. A maximum requires $A'(w)=0$, so the stated dimensions are not assumed optimal without that check.
=> FINAL: $A(w)=w(10-w)$; $A(7)=21$; check $A'(w)=0$ (check units/interpretation)

Question CALC1-FINAL-Q04
scratch: show step
1. $x^2-25=(x-5)(x+5)$
2. cancel $x-a$ only after identifying $x\neq a$
3. $\lim_{x\to 5} x+5=10$
=> FINAL: 10 (check units/interpretation)

Question CALC1-FINAL-Q05
scratch: show step
1. Critical point: $f'(c)=0$ or $f'(c)$ is undefined (when f is defined).
2. A sign change in f' from + to – indicates a local maximum.
=> FINAL: critical point: $f'(c)=0$ or undefined; + to – sign change gives a local maximum (check units/interpretation)

Question CALC1-FINAL-Q06
scratch: show step
1. $f'(x)=2x(3x-1)+(x^2+4)3$
2. $f'(1)=2(2)+(5)3=11$
=> FINAL: 11 (check units/interpretation)

Question CALC1-FINAL-Q07
scratch: show step
1. $\int 3x\,dx = (3/2)x^2$

2. [3/2 x^2]_0^2 = 6
=> FINAL: 6 (check units/interpretation)

Question CALC1-FINAL-Q08
scratch: show step
1. 2w+2l=14 ⇒ l=7-w
2. A(w)=w(7-w); at w=4, A=12
3. A maximum requires A'(w)=0, so the stated dimensions are not assumed optimal without that check.
=> FINAL: A(w)=w(7-w); A(4)=12; check A'(w)=0 (check units/interpretation)

Question CALC1-FINAL-Q09
scratch: show step
1. x^2-25=(x-5)(x+5)
2. cancel x-a only after identifying x≠a
3. lim_(x→5) x+5=10
=> FINAL: 10 (check units/interpretation)

Question CALC1-FINAL-Q10
scratch: show step
1. Critical point: f'(c)=0 or f'(c) is undefined (when f is defined).
2. A sign change in f' from + to – indicates a local maximum.
=> FINAL: critical point: f'(c)=0 or undefined; + to – sign change gives a local maximum (check units/interpretation)